

Submission Worksheet

Submission Data

Course: IT202-008-S2026

Assignment: IT202 Module 4 SQL Challenge

Student: Angel V. (av847)

Status: Submitted | **Worksheet Progress:** 100%

Potential Grade: 10.00/10.00 (100.00%)

Received Grade: 0.00/10.00 (0.00%)

Started: 3/22/2026 10:28:05 PM

Updated: 3/25/2026 2:11:44 AM

Grading Link: <https://learn.ethereallab.app/assignment/v3/IT202-008-S2026/it202-module-4-sql-challenge/grading/av847>

View Link: <https://learn.ethereallab.app/assignment/v3/IT202-008-S2026/it202-module-4-sql-challenge/view/av847>

Instructions

- Overview Link: <https://youtu.be/oW7AK39mgmo>

1. Ensure you read all instructions and objectives before starting.

2. Create a new branch from qa called M4-Homework

1. `git checkout qa` (ensure proper starting branch)
2. `git pull origin qa` (ensure history is up to date)
3. `git checkout -b M4-Homework` (create and switch to branch)

3. Copy the template code from here: [GitHub Repository - M4 Homework](#)

- Put all into an M4 folder inside your `public_html`

- Expected files

- `sql/000_create_table_todos.sql`
- `sql/init_db.php`
- `todos/create.php`
- `todos/pending.php`
- `todos/completed.php`
- `nav.php`
- two index files (M4 and todos)

- Immediately record to history

- `git add public_html`
- `git commit -m "adding M4 HW baseline files"`
- `git push origin M4-Homework`
- Create a Pull Request from M4-Homework to qa and keep it open

4. Fill out the below worksheet

- Each Problem requires the following as you work

- Ensure there's a comment with your UCID, date, and brief summary of how the problem was solved
- Update ucid in `nav.php` where noted
- Code solution (add/commit periodically as needed) (refer to comments in the create, pending,

completed files)

5. Once finished, click "Submit and Export"
6. Locally add the generated PDF to a folder of your choosing inside your repository folder and move it to Github
 1. `git add .`
 2. `git commit -m "adding PDF"`
 3. `git push origin M4-Homework`
 4. On Github merge the pull request from M4-Homework to qa
 5. On Github create a pull request from qa to prod and immediately merge. (This will trigger the prod deploy to make the render.com prod links work)
7. Upload the same PDF to Canvas
8. Sync Local
 1. `git checkout qa`
 2. `git pull origin qa`

Section #1: (1 pt.) Prep

Progress: 100%

Details:

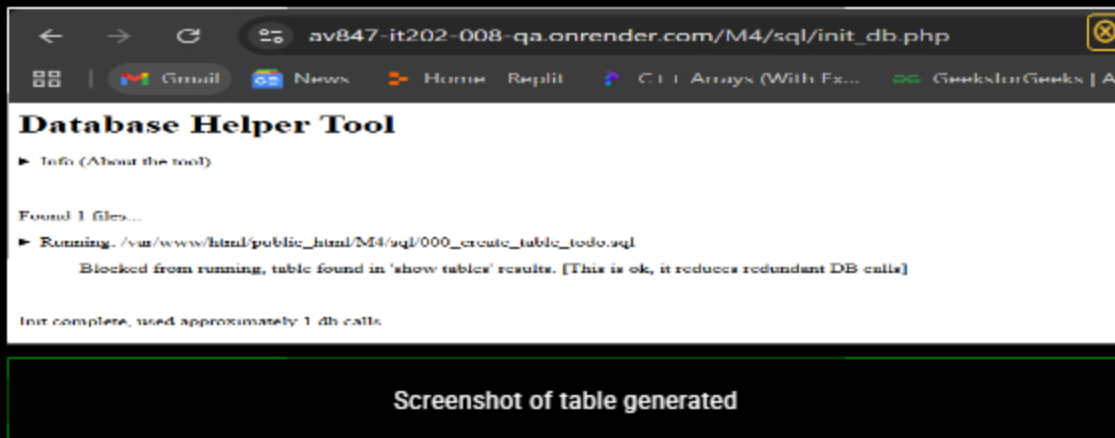
Once the files are copied in, run your local PHP server and visit `/M4/sql/init_db.php` in the browser to trigger the table generation.

Task #1 (0.50 pts.) - Show the output of the init_db.php tool

Progress: 100%

Details:

The output should note either successfully executed or skipped (if you ran this already)



Database Helper Tool

► Info (About the tool)

Found 1 files...

► Running: `/var/www/html/public_html/M4/sql/000_create_table_todo.sql`

Blocked from running, table found in 'show tables' results. [This is ok, it reduces redundant DB calls]

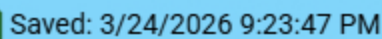
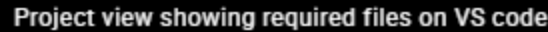
Init complete, used approximately 1 db calls

Screenshot of table generated

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Task #2 (0.50 pts.) - Show the project view of your editor to show all files are present

Progress: 100%



Progress: 80%

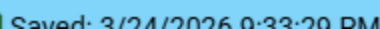
Progress: 100%

Progress: 100%

Show the code snippet related to the completed HTML form

code snippet of completed HTML form (part 1)

code snippet of completed HTML form (part 2)



Part 2:

Progress: 100%

Details:
Items

- Note the field names and why the specific types were chosen
- Also note how you made "self" the default for the assigned input

Your Response:

The field names that were chosen come directly from the \$expected_fields array that was pre defined. The "self" default was set with the "value" attribute using the null coalescing operator.

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Task #2 (0.75 pts.) - Form Submission PHP Validations

Progress: 100%

Part 1:

Progress: 100%

Details:

- Validate the incoming data for correct format based on the SQL table definition.
- When not valid, provide a user-friendly message of what specifically was wrong and set \$is_valid to false
- Assigned should check for "self" if a valid format/value isn't provided; it should default to "self" if all checks for this property fail
- Include screenshot(s) from Render.com QA demonstrating each validation (ensure URL is visible)

```
// VALIDATE THE INCOMING DATA FOR CORRECT FORMAT BASED ON THE DB TABLE DEFINITION
// WHEN NOT VALID, PROVIDE A USER-FRIENDLY MESSAGE OF WHAT SPECIFICALLY WAS WRONG AND SET $is_valid TO FALSE
// ASSIGNED SHOULD CHECK FOR "self" IF A VALID FORMAT/VALUE ISN'T PROVIDED; IT SHOULD DEFAULT TO "self" IF ALL CHECKS FOR THIS PROPERTY FAIL
// INCLUDE SCREENSHOT(S) FROM RENDER.COM QA DEMONSTRATING EACH VALIDATION (ENSURE URL IS VISIBLE)

// Task Description Validation
if (isset($_POST['task_description']) && !empty($_POST['task_description'])) {
    $task_description = $_POST['task_description'];
    // Check if the task description is empty
    if (trim($task_description) === '') {
        $is_valid = false;
        $user_message[] = "Task description is empty. Please fill out description.";
    }
    // Check if the task description is less than 128 characters
    if (strlen($task_description) > 128) {
        $is_valid = false;
        $user_message[] = "Task description must be less than 128 characters.";
    }
}

// URL Validation
if (isset($_POST['url']) && !empty($_POST['url'])) {
    $url = $_POST['url'];
    // Check if the URL is a valid format
    if (!filter_var($url, FILTER_VALIDATE_URL)) {
        $is_valid = false;
        $user_message[] = "The URL data is not in a valid URL format (YYYY-MM-DD).";
    }
}

// Assigned Validation
if (isset($_POST['assigned']) && !empty($_POST['assigned'])) {
    $assigned = $_POST['assigned'];
    // Check if the assigned value is not empty
    if (trim($assigned) === '') {
        $is_valid = false;
        $user_message[] = "The assigned value must be 40 characters or fewer.";
    }
}

// End validation logic
```

PHP validations solved

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Part 2:

Details:

Briefly explain your validation logic.

Your Response:

First, I checked and made sure "task" is validated with the empty(trim()) function. Along with that I used the strlen() function to enforce the column limit of 128. Next I needed to check and validate "due" and I did that using "DateTime::createFromFormat()" function which compared the reformatted result with the original input. Lastly, I checked and validated "assigned" using the strlen() against the limit of 60 characters. If any check fails then \$user_message array is prompted and the \$is_valid variable is set to false.



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≡ Task #3 (0.75 pts.) - Show the code snippet related to the INSERT query

Progress: 100%

Part 1:

Progress: 100%

Details:

Design a query to insert the incoming data to the proper columns. Ensure valid and proper PDO named placeholders are used. <https://phpdelusions.net/pdo>

```
$query = "INSERT INTO M4_TODO5 (task, due, assigned) VALUES(:task, :due, :assigned)";
$params = [ ":task" => trim($task), ":due" => $due, ":assigned" => trim($assigned) ]; // Apply the proper PDO placeholders
try {
    $db = getDB();
    $stmt = $db->prepare($query);
    $r = $stmt->execute($params);
    if ($r) {
        echo "Inserted new Todo with id " . $db->lastInsertId();
    } else {
        echo "Failed to insert";
    }
}
```

Screenshot of INSERT query



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Part 2:

Progress: 100%

Details:

Briefly describe how the query works in plain words.

Your Response:

The INSERT query first inserts a new row into M4_TODOS by supplying values for the task,due, and assigned fields. The then remaining columns are left to their schema defaults id auto increments and "is_complete defaults to 0. Completed then defaults to NULL and created/modified both default to the current timestamp.



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☰ Task #4 (+ 1.01 pts.) - Extra Credit: Handle Unique Constraint Errors

Progress: 0%

📁 Part 1:

Progress: 0%

Details:

Show the code related to handling the unique constraint errors and the user-friendly message



Missing Caption



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☞ Part 2:

Progress: 0%

Details:

Briefly explain how your solution works. Describe the logic, steps, or decisions your code makes to reach the result. Do not restate the problem or list requirements.

Your Response:

Missing Response



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☞ Task #5 (0.75 pts.) - Create Page URLs

Progress: 100%

Details:

- Direct link to the file in the homework-related branch from Github (should end in .php, , zero credit if it does not)
- Direct link to the file on Render.com Prod (Grab the qa url you're testing, paste it, change "-qa" to "-prod"), zero credit if not prod

URL #1

https://github.com/av847-NJIT/av847-IT202-008/blob/M4-Homework/public_html/M4/todos/create.php



URL

https://github.com/av847-NJIT/av847-IT202-008/blob/M4-Homework/public_html/M4/todos/create.php



URL #2

<https://av847-it202-008-prod.onrender.com/M4/todos/create.php>



URL

<https://av847-it202-008-prod.onrender.com/M4/todos/create.php>



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Section #3: (3 pts.) Pending Page

Progress: 100%

Task #1 (1 pt.) - Fetching Todos

Progress: 100%

Part 1:

Progress: 100%

Details:

- Refer to the HTML table in the given code and build a query that'll select the columns in the same order as the table from the Todo table
- Cross-reference the HTML table columns with what'd most plausibly match the SQL table, aside from the notes below
- For the Status part, you'll need to calculate the "days_offset" from the due date, ensure the virtual column matches "days_offset"
- For Actions, this isn't part of the query, and there's nothing special to select for it
- Filter the results where the todo item is NOT completed and order the results by those due soonest
- No limit is required
- Include screenshot(s) from Render.com QA demonstrating 5 different records (ensure URL is visible)



ID	Task	Due Date	Status	Assigned	Actions
9	Buy groceries	2024-03-27	Due in 2 day(s)	self	Complete
11	Watch exercise class	2024-03-27	Due in 2 day(s)	self	Complete
7	go to dentist	2024-03-28	Due in 3 day(s)	self	Complete
8	Go for run	2024-03-28	Due in 3 day(s)	self	Complete
10	Buy mom's gift	2024-03-29	Due in 3 day(s)	self	Complete

Screenshot of 5 different records

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Part 2:

Progress: 100%

Details:

Briefly describe how the query works in plain words.

Your Response:

In the Update query, I set it to where a single todo is marked complete when it is set to 1. The record is also recorded at that point. Next, the current timestamp is marked as the completion date only if the it isn't already complete.

In the the Select query, I set it to fetch all incomplete todos and calculate how many days until the due date. Next, I sort them so the most urgent items appear first.

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Task #2 (1 pt.) - Completing Todos

Progress: 100%

Part 1:

Progress: 100%

Details:

- Create a query that'll update the respective ToDo, marking it complete and setting the date for the completed date field as today
- Ensure the "id" is utilized using proper PDO named placeholders so that only the one item is updated
- Add an extra clause to update only if the complete field of the record is not set

av847-it202-008-ga.onrender.com/1M4/todos/completed.php					
Gmail News Home - Repl.it C++ Arrays (With Ex... GecktorGeeks A C... Learn C and C++ Pr... W3Schools Online... vidoc.pub_computer...					
Create Todos Pending Todos Completed Todos					
av847 Wednesday, March 25, 2026 at 1:18 AM					
Completed Todos					
ID	Task	Due Date	Completed Date	Status	Assigned
5	Go to mechanic	2026-03-28	2026-03-25	Completed 3 day(s)	self
4	Go to Doctor	2026-03-27	2026-03-25	Completed 2 day(s)	self
1	workout	2026-03-29	2026-03-25	Completed 4 day(s)	self
2	Walk dog	2026-03-25	2026-03-25	Completed 0 day(s)	self
3	Homework	2026-03-25	2026-03-25	Completed 0 day(s)	self



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Part 2:

Progress: 100%

Details:

Briefly describe how the query works in plain words.

Your Response:

The query works by fetching all completed todos and first extracting the date portion from the completed timestamp. Next, it calculates how many days early or late the task was finished relative to its due date. The result is then sorted by most recently completed first followed by most recently due.



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Task #3 (1 pt.) - Pending Page URLs

Progress: 100%

Details:

- Direct link to the file in the homework-related branch from GitHub (should end in .php, , zero credit if it does not)
- Direct link to the file on Render.com Prod (Grab the qa url you're testing, paste it, change "-qa" to "-prod"), zero credit if not prod

URL #1

https://github.com/av847-NJIT/av847-IT202-008/blob/M4-Homework/public_html/M4/todos/pending.php



URL

<https://github.com/av847-NJ>**URL #2**

<https://av847-it202-008-prod.onrender.com/M4/todos/pending.php>



URL

<https://av847-it202-008-prod.>

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Section #4: (2 pts.) Completed Page

Progress: 100%

Task #1 (1 pt.) - Fetching Todos

Progress: 100%

Details:

- Refer to the HTML table below and build a query that'll select the columns in the same order as the table from the Todo table
- Cross-reference the HTML table columns with what'd most plausibly match the SQL table, aside from the notes below
- For the completed date, you'll need to extract the date portion from the completed column
- For the Status part, you'll need to calculate the "days_offset" from the completed date, ensure the virtual column matches "days_offset"
- Filter the results where the todo item is completed and order the results by most recently completed and most recently due
- No limit is required
- Include screenshot(s) from Render.com QA demonstrating 5 different records (ensure URL is visible)

av547-1t202-006-qa.onrender.com/M4/todos/completed.php

av547 | Wednesday, March 23, 2026 at 1:49 AM

Completed Todos

ID	Task	Due Date	Completed Date	Status	Assigned
5	Go to market	2026-04-18	2026-04-15	Completed 3 days(s)	self
7	go to dentist	2026-04-18	2026-04-15	Completed 3 days(s)	self
8	Go for run	2026-04-18	2026-04-15	Completed 3 days(s)	self
10	Buy cousin's gift	2026-04-28	2026-04-25	Completed 3 days(s)	self
4	Go to doctor	2026-04-27	2026-04-25	Completed 2 days(s)	self
9	Buy groceries	2026-04-27	2026-04-25	Completed 2 days(s)	self
11	Watch favorite show	2026-04-27	2026-04-25	Completed 2 days(s)	self
1	workout	2026-04-15	2026-04-15	Completed 0 days(s)	self
3	Walk dog	2026-04-15	2026-04-15	Completed 0 days(s)	self
2	Housework	2026-04-15	2026-04-15	Completed 0 days(s)	self

Screenshot of previous 5 records completed and added to table

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Part 2:

Details:

Briefly describe how the query works in plain words.

Your Response:

Already described above: The query works by fetching all completed todos and first extracting the date portion from the completed timestamp. Next, it calculates how many days early or late the task was finished relative to its due date. The result is then sorted by most recently completed first followed by most recently due.

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Details:

- Direct link to the file in the homework-related branch from GitHub (should end in .php, , zero credit if it does not)
- Direct link to the file on Render.com Prod (Grab the qa url you're testing, paste it, change "-qa" to "-prod"), zero credit if not prod

URL #1

https://github.com/av847-NJIT/av847-IT202-008/blob/M4-Homework/public_html/M4/todos/completed.php



URL

https://github.com/av847-NJIT/av847-IT202-008/blob/M4-Homework/public_html/M4/todos/completed.php
**URL #2**

<https://av847-it202-008-prod.onrender.com/M4/todos/completed.php>



URL

<https://av847-it202-008-prod.onrender.com/M4/todos/completed.php>


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Section #5: (1 pt.) Misc

Progress: 100%

Task #1 (0.33 pts.) - Github Details

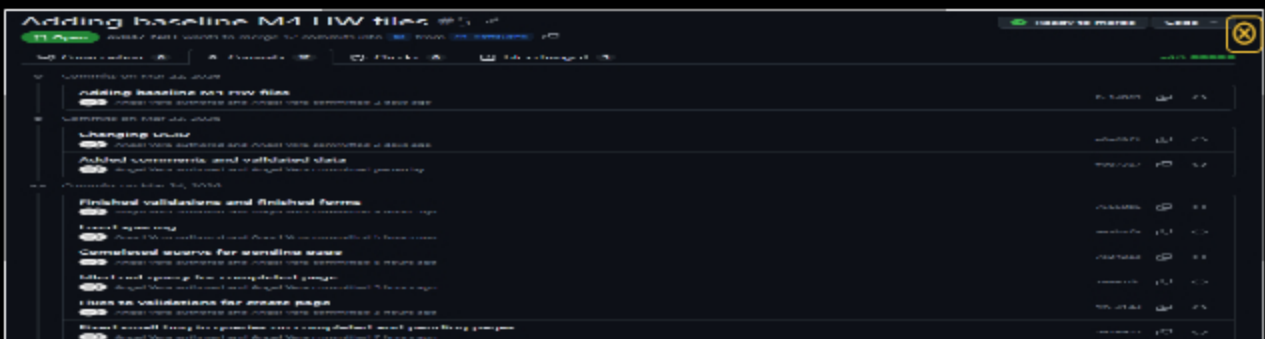
Progress: 100%

Part 1:

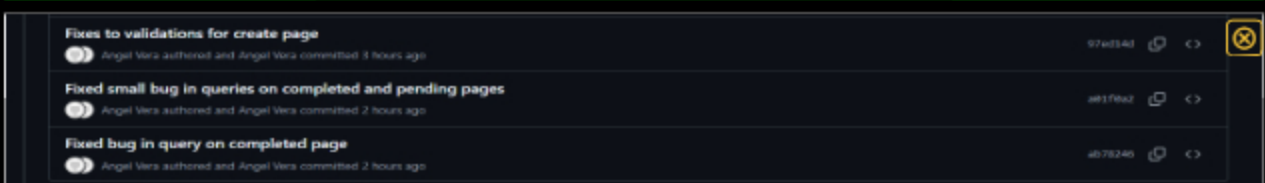
Progress: 100%

Details:

From the Commits tab of the Pull Request screenshot the commit history



Screenshot of commit history(part 1)



Commits on Mar 25, 2026	
Fixes to validations in create.php Angel Vera authored and Angel Vera committed 1 hour ago	931b08c
Fixed bugs in create/pending files and fixed bug in in create file Angel Vera authored and Angel Vera committed 1 hour ago	53104aa

Screenshot of commit history(part 2)

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Part 2:

Progress: 100%

Details:

Include the link to the Pull Request (should end in /pull/#)

URL #1

<https://github.com/av847-NJIT/av847-IT202-008>



URL

<https://github.com/av847-NJIT/av847-IT202-008>

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Task #2 (0.33 pts.) - WakaTime - Activity

Progress: 100%

Details:

- Visit the WakaTime.com Dashboard
- Click Projects and find your repository
- Capture the overall time at the top that includes the repository name
- Capture the individual time at the bottom that includes the file time
- Note: The duration isn't relevant for the grade and the visual graphs aren't necessary

Projects • av847-IT202-008

3 hrs 46 mins over the Last 7 Days in av847-IT202-008 under all branches.

Overview

Files

1 hr 51 mins	M4/todos/create.php
36 mins	M4/todos/completed.php
27 mins	M4/todos/pending.php
	M4/sql/000_create_table_todo.sql

Branches

3 hrs 46 mins M4-Homework

25 mins M4/nav.php
6 mins M4/sql/init_db.php
5 mins M4/todos/index.php
35 secs M4/sql/README.md
4 secs M4/index.php
1 sec M3/challenge3.html
1 sec M3/challenge2.html
0 secs M3/challenge1.html

Individual file overview



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≡ Task #3 (0.33 pts.) - Reflection

Progress: 100%

⇒ Task #1 (0.33 pts.) - What did you learn?

Progress: 100%

Details:

Briefly answer the question (at least a few decent sentences)

Your Response:

What I learned during this assignment was how to use a database sql table and successfully transfer data into it. I also learned how to properly and successfully fill out forms in html. I also learned how to fill out queries and get them to manipulate the data in the way I want.



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⇒ Task #2 (0.33 pts.) - What was the easiest part of the assignment?

Progress: 100%

Details:

Briefly answer the question (at least a few decent sentences)

Your Response:

The easiest part of the assignment was copying and pasting the files. There wasn't many things I found easy in this assignment but through research I was able to figure out how everything works.



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≡ Task #3 (0.33 pts.) - What was the hardest part of the assignment?

Progress: 100%

Details:

Briefly answer the question (at least a few decent sentences)

Your Response:

The hardest part of the assignment was first making sure the data was properly submitting into the table. It took me quite a while to debug that aspect of the assignment and get it running properly. Once finished I also found it difficult debugging as I had a few small syntax errors in a couple of files that took me a while to clean up.



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